

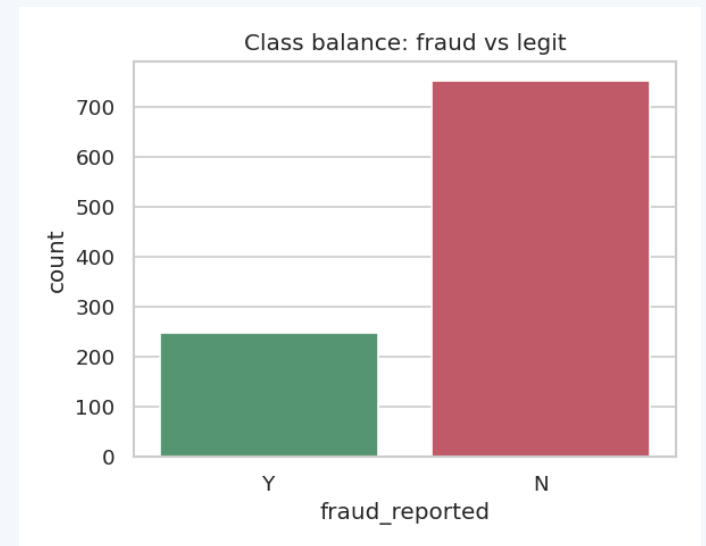
Micro Insurance Claim Risk & Anomaly Detection System

Capstone Project — Himshikhar AIML Track — Project 10

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- Insurance teams cannot manually inspect every claim in detail.
- Most claims are genuine; a minority are fraudulent.
- Manual review of everything is slow and expensive.
- Our system flags suspicious / abnormal claims for human review.
- Impact: reviewers focus time on the riskiest claims; genuine claims
- move through a fast track — reducing cost and operational risk.

- Insurance Fraud Detection (Kaggle) — auto-insurance claims.
- 1,000 claims × 39 columns.
- Target: fraud_reported (Y / N).
- Fraud rate: 24.7% — imbalanced (most claims legitimate).
- '?' is the dataset's missing-value token — cleaned in preprocessing.





Two models run in parallel: a supervised classifier + an anomaly detector.

- 1. Supervised classifier — Random Forest (class-weight balanced)
 - outputs a fraud probability for each claim.
- 2. Anomaly detector — Isolation Forest flags unusual claims
 - (unsupervised safety net, catches novel patterns).
- 3. Reason codes — transparent rules explain each flag in plain English.
- 4. Recommendation engine — combines score + anomaly into one action:
 - Auto-approve / Manual review / High-priority review.

- Interactive app scores a single claim or a whole CSV.
- Shows fraud probability, anomaly flag, reason codes,
- and a colour-coded recommendation for the reviewer.

App screenshot placeholder

[Add your app screenshots here for the final submission]

0.769

ROC-AUC

0.64

Precision (fraud)

0.53

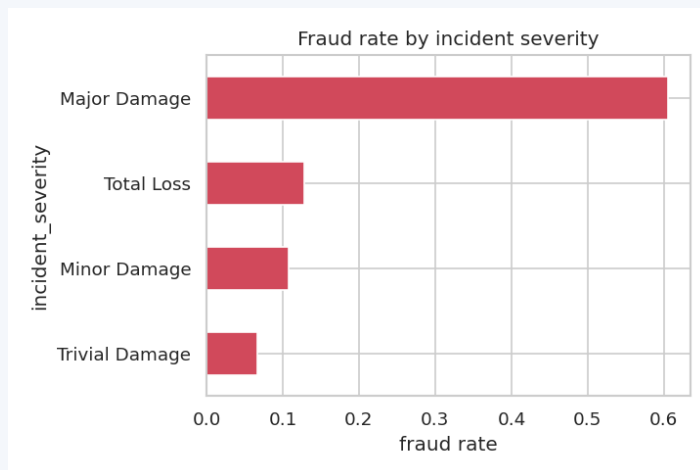
Recall (fraud)

0.58

F1 (fraud)

Top fraud signals:

1. Incident severity = Major Damage
2. Vehicle / property / injury amounts
3. Claim-to-premium ratio
4. Total claim amount



- Small, single-line dataset (1,000 auto claims only).
- Historical labels may carry human bias.
- ~53% fraud recall — some fraud is still missed.
- Reason codes are illustrative rules, not full causal proof.
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- Responsible use: DECISION-SUPPORT ONLY.
- The system never auto-rejects a claim — every flag goes to a human,
- and the final decision always rests with a trained claims officer.

- Gradient boosting (XGBoost / LightGBM) with probability calibration.
- SHAP values for per-claim explanations (beyond rule-based reasons).
- Threshold tuning to balance recall vs. reviewer workload.
- Larger, multi-line dataset with periodic retraining.
- Drift monitoring to detect changing fraud patterns over time.

Conclusion

A working, end-to-end system that scores claims for fraud risk,
flags anomalies, explains why, and recommends a clear action.
Backend (models) + frontend (Streamlit app) + documentation, all delivered.
Built responsibly as a decision-support tool for human reviewers.